

**Faculté des HEC
Université de Lausanne**

Master of Science in Finance

SUSTAINABILITY AWARE ASSET MANAGEMENT

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General comments on Projects

- Sharpe Ratio
- Cumulative Returns
- Value-Weighted Portfolio
- Sample
- Plots of cumulative return of value-weighted portfolios

Risk-Adjusted Return

Sharpe Ratio

The Sharpe ratio measures the return of a strategy per unit of risk (i.e., the volatility):

$$SR_p = \frac{\bar{R}_p - R_f}{\sigma_p}$$

The Sharpe ratio is usually computed ex post, at the end of the investment period, to assess the performance of the strategy.

R_f should be defined with the same convention as $\bar{R}_p$

Example:

If you compute $\bar{R}_p$ as the annualized average return (say, 0.06 for 6%)

Then you should take the average risk-free rate over the sample (say, 0.015 for 1.5%)

Cumulative Returns

We have cumulated monthly returns over T periods (i.e., $T/12$ years).

- If we use simple returns, the cumulative return is

$$R_p[T] = \prod_{k=1}^T (1 + R_{p,t+i}) - 1$$

The annualized simple return over the sample is given by

$$\bar{R}_p^{(y)} = (1 + R_p[T])^{12/T} - 1$$

- $R_{p,t+i}$ should be monthly returns
- Report cumulative returns of value-weighted and min-variance portfolios on the same plot
- Start the plot at 1

Value-Weighted Portfolio

The weights of the value-weighted portfolio are $\alpha_{i,t}^{(vw)} = Cap_{i,t} / \sum_{j=1}^N Cap_{j,t}$ where $Cap_{i,t}$ is the market capitalization at the end of period t . The portfolio return is given by

$$R_{t+1}^{(vw)} = \sum_{i=1}^N \alpha_{i,t}^{(vw)} R_{i,t+1}$$

Use the weights from the previous period ($\alpha_{i,t}^{(vw)}$ for computing $R_{t+1}^{(vw)}$).

Sample

The first allocation should take place on December 2013

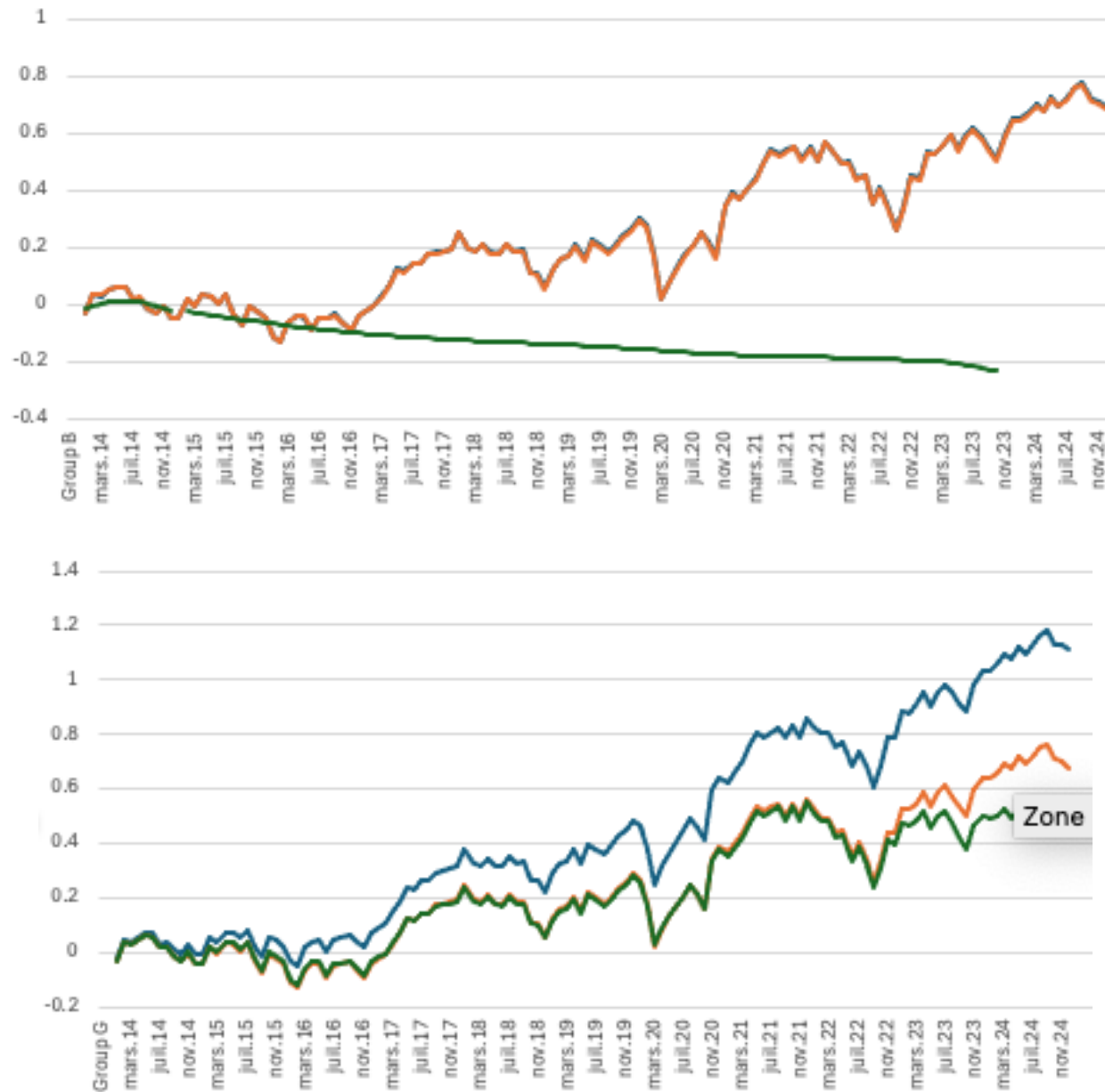
The last allocation should take place on December 2023

Monthly returns can be computed for 2024 by updating weights

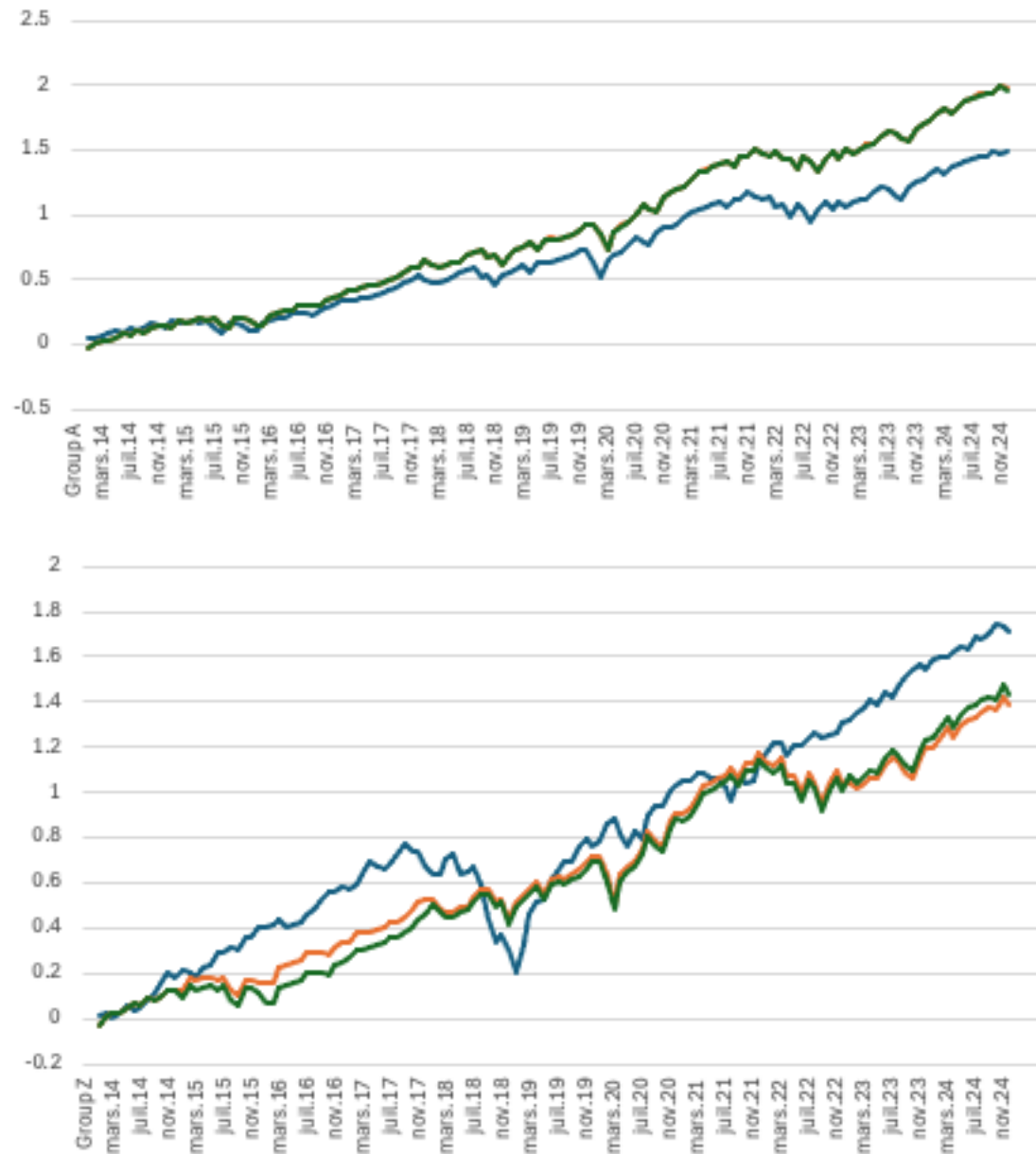
So, plots should cover the period from January 2014 to December 2024

(or performances are not comparable)

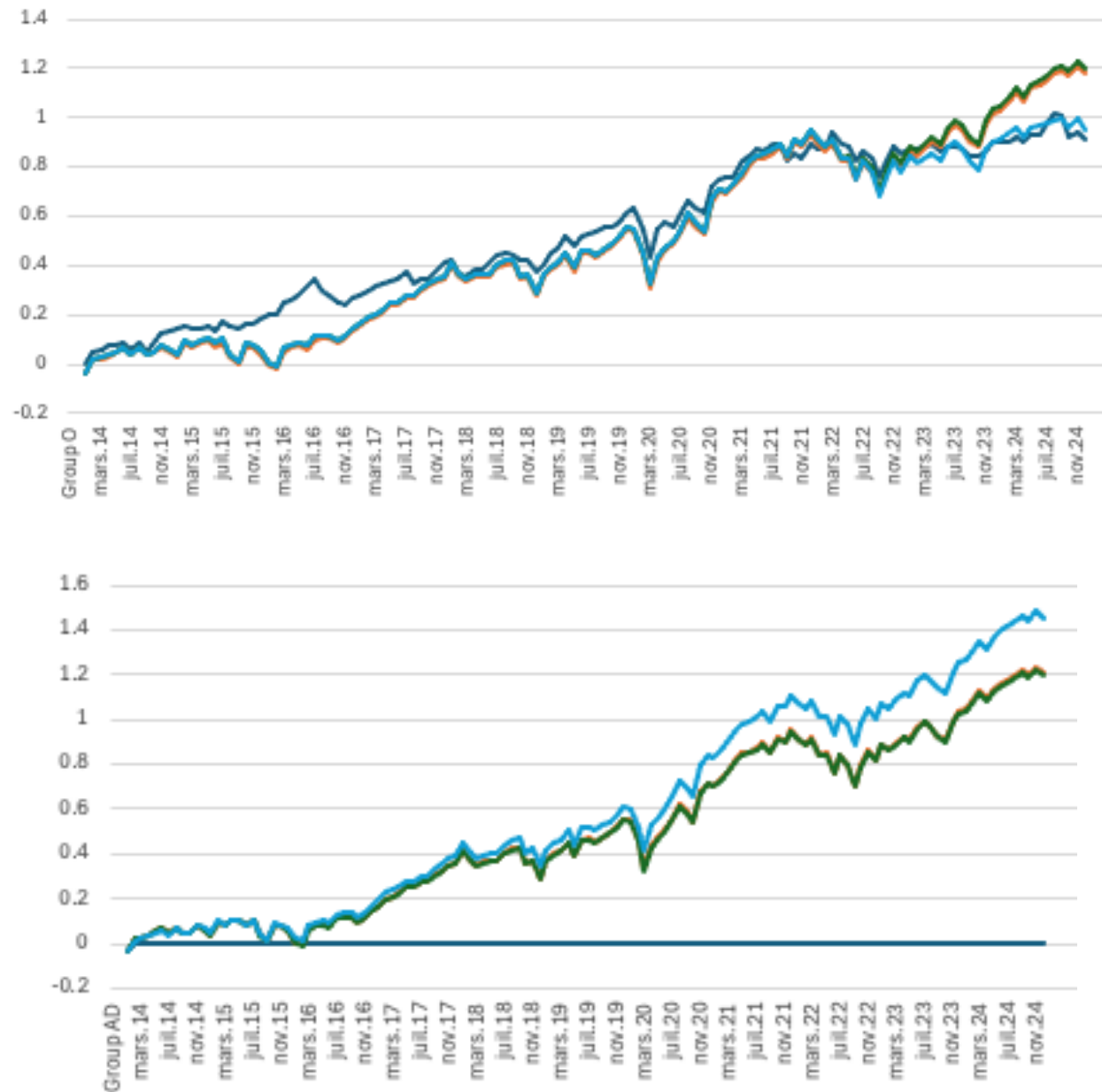
Consistency Cumulative returns – Europe



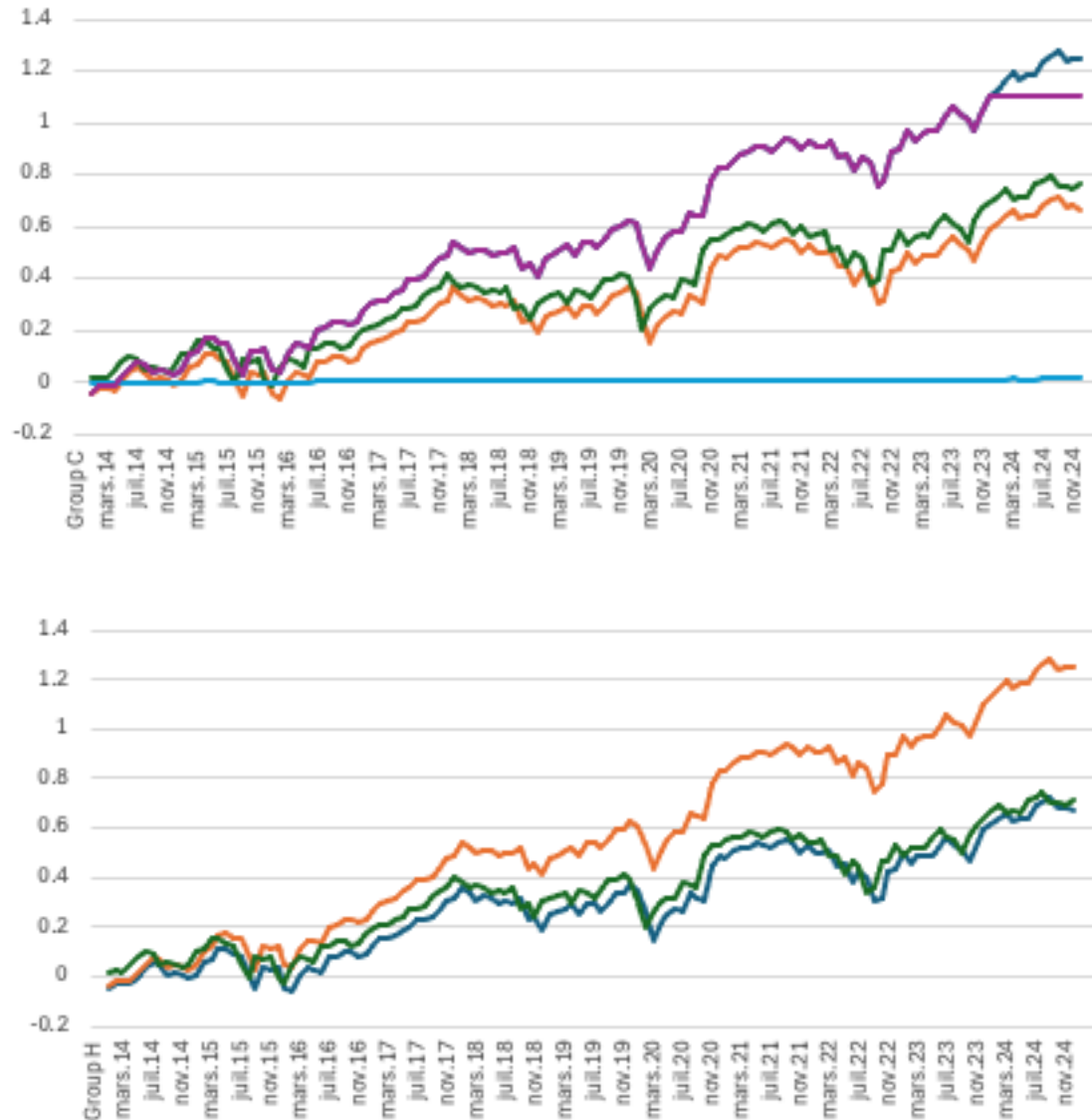
Consistency Cumulative returns – North America



Consistency Cumulative returns – North America-Europe



Consistency Cumulative returns – Pacific



Consistency Cumulative returns – Emerging markets

